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AN APPROACH TOWARDS BUILDING A SMART LIFT SYSTEM USING MACHINE LEARNING

PROJECT GUIDE
DR. SUDHIR DHAGE

AT PRESENT...

- Lift systems heavily rely on the pre-programmed behaviour
- Lack of real-time adaptability to user patterns
- Increased energy consumption due to static operation
- Longer wait times for users during peak demand periods
- Buildings with multiple lifts only worsen the energy situation
- Difficulty in optimizing routes and energy usage

PROBLEM DEFINITION

In modern buildings, lift systems are essential for efficient transportation across floors. However, conventional lift systems often operate on basic pre-programmed logic, leading to inefficiencies such as prolonged wait times, unnecessary stops, and high energy consumption. This project aims to develop a smart lift system using machine learning techniques to optimize lift operations. By predicting future lift calls based on historical usage patterns and current conditions, the system will improve the overall efficiency, reducing wait times and energy consumption. The model will be trained on synthesized time-series data to anticipate user demands and optimize lift movement.

- How to make the lift system ‘know’ where the next request is going to be from? - Using RNNs with Time-Series Data

TO BE DISCUSSED...

- Scope and Objectives
- Literature Review
- General Interaction Model
- Proposed Simulation Model
- Initial Experimentation with a Single Lift
- Approach Towards Multi-Lift Systems
- Results
- Conclusion
- Further work
- References

SCOPE

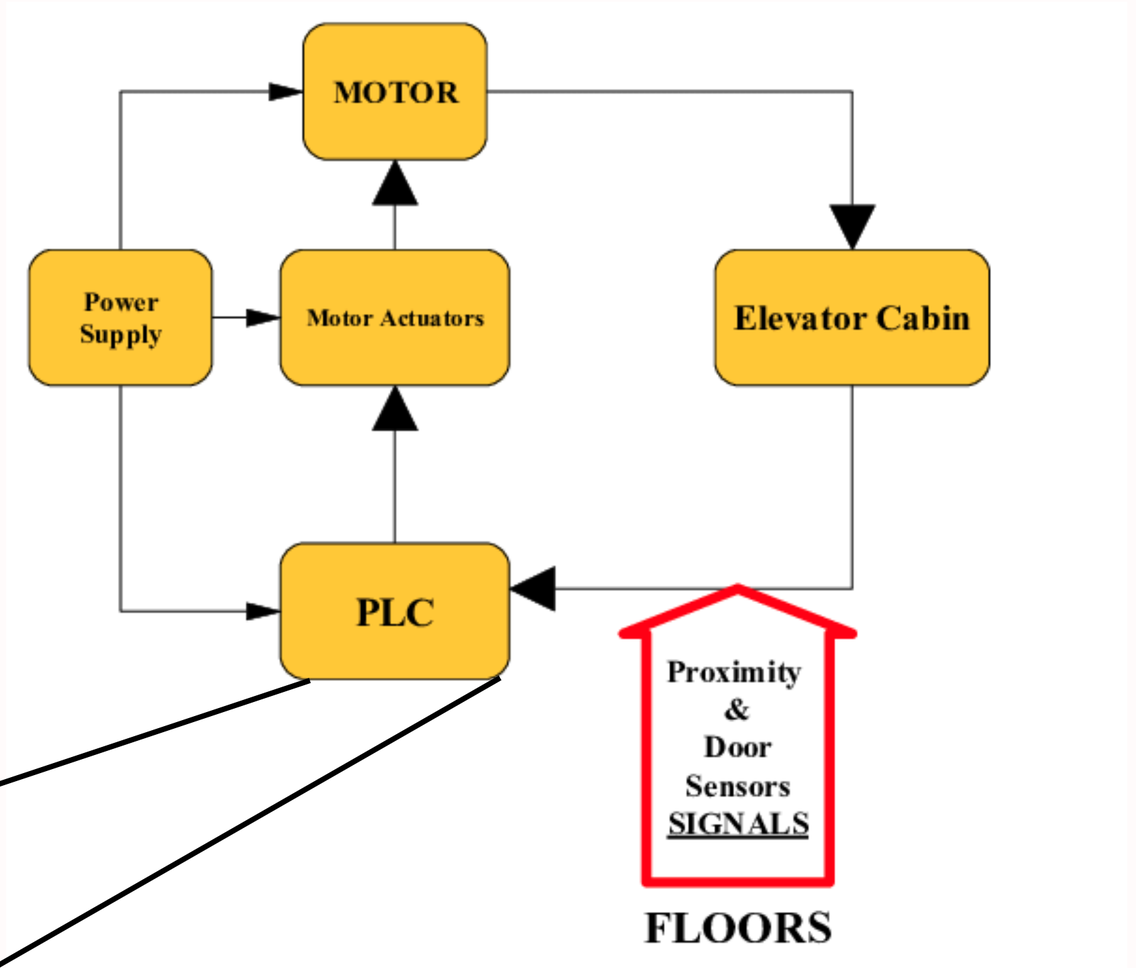
- Under consideration - Residential lifts, Office Building lifts.
- Working with synthetic time-series data to simulate lift usage for model training, followed by training on Real Data.
- Build and train predictive models to forecast future lift calls and optimize movement.
- Design and develop a dedicated simulation software to simulate various lift models for boosting the research.
- Provide software-based Solutions.

PROJECT OBJECTIVES

- To enhance user experience by minimizing wait times for the lift.
- To design lift systems that optimize power consumption and reduce overall energy usage.
- To predict lift usage patterns using machine learning for efficient routing and operations.
- To develop a scalable system adaptable to different building sizes and usage scenarios.
- To reduce unnecessary lift stops and improve the overall efficiency of lift movement.
- To integrate data-driven decision-making for smarter, real-time lift operations.

LITERATURE REVIEW

- Current lift systems manage lift calls through PLC (Programmable Logic Controller)
- PLC is coded and followed as a protocol
- Introducing a ‘smart’ component over the PLC may help optimize power usage and reduce wait times



Current Lift System Schematic

POWER CALCULATIONS

2 ESTIMATED DAILY ENERGY CONSUMPTION

The estimated daily energy consumption (E_d) of a lift is the sum of the running consumption (E_{rd}) and the standing (idle/standby) consumption (E_{sd}):

$$E_d = E_{rd} + E_{sd} \quad \ldots (1)$$

$$E_{rd} = \frac{n_d \times \%S \times k_L \times E_{rc}}{2} \quad \ldots (2)$$

where:

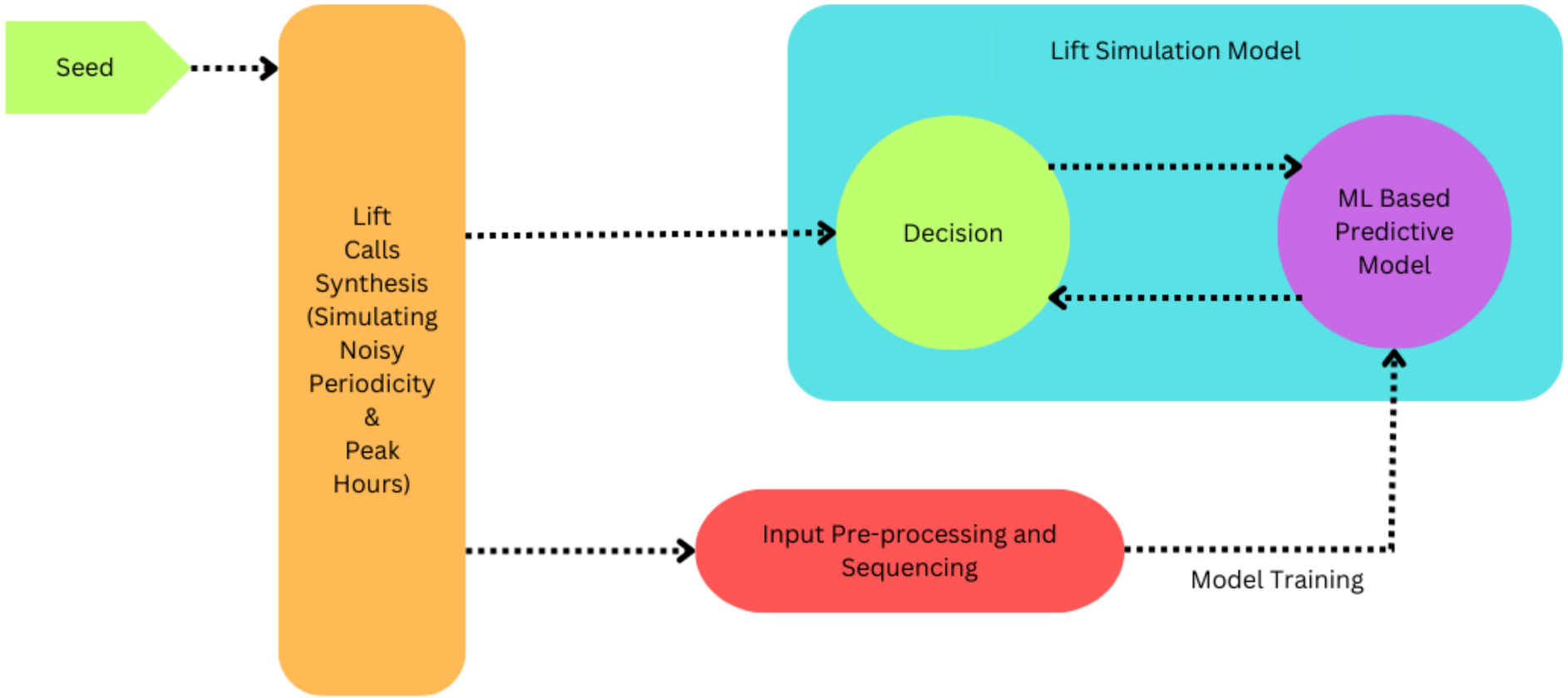
n_d is the number of trips per day[4]
$\%S$ is the percentage average travel distance per trip for a target installation
k_L is the load factor per trip
E_{rc} is the measured or estimated running energy consumption of the ISO reference cycle (two trips) in Wh.

$$E_{sd} = (24 - \frac{n_d}{3600} t_{av})(P_{id} R_{id} + P_{st} R_{st}) \quad \ldots (4)$$

where:

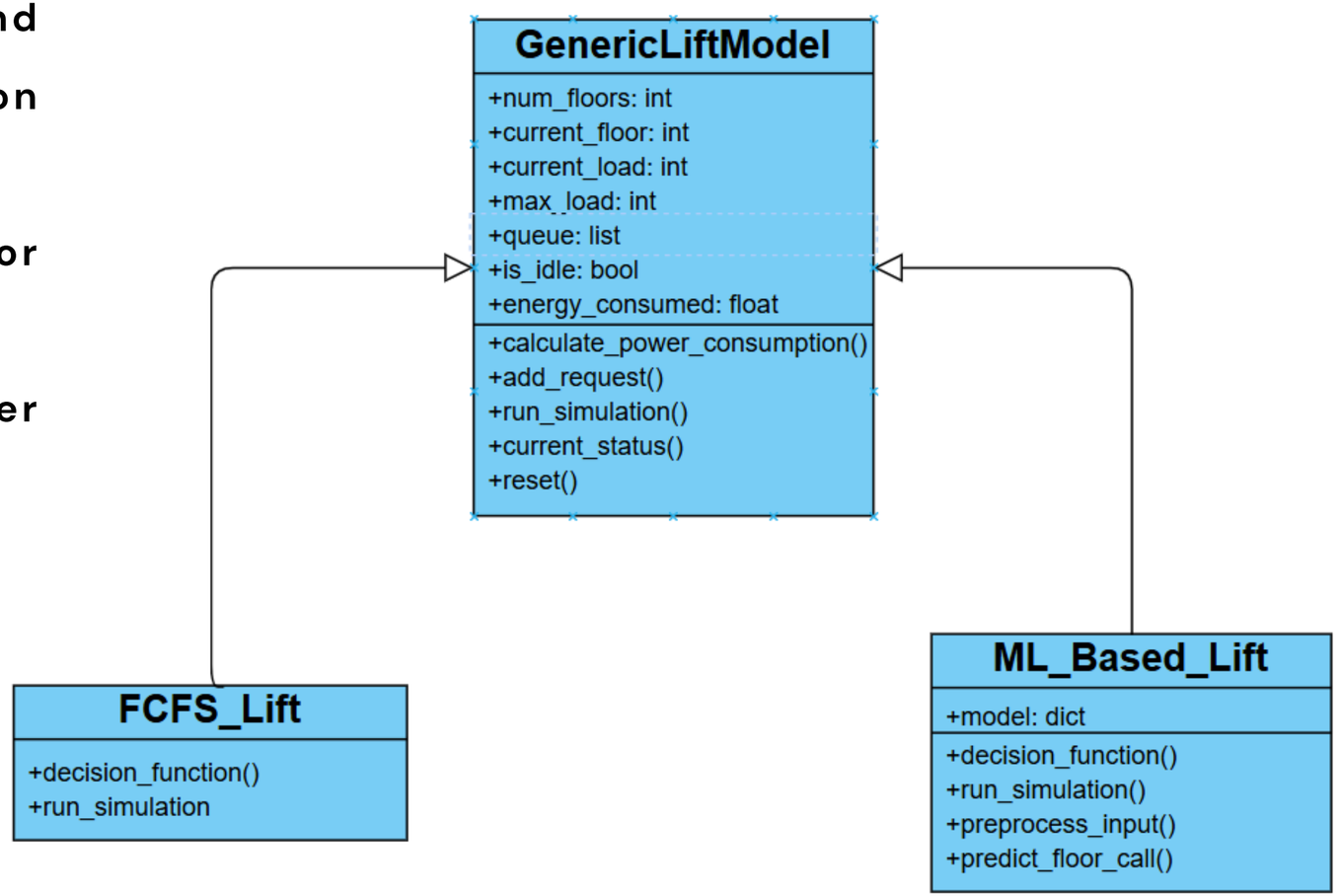
P_{id} is the power used when the lift is in idle mode (W)
(measured after door operations have ceased when stopped at a landing.)
P_{st} is the power used when the lift is in standby mode (W)
(measured after 5 minutes of inactivity.)
R_{id} is the ratio of idle time when consuming P_{id} (value<1)
R_{st} is the ratio of standby time when consuming P_{st} (value<1)
t_{av} is the time to travel the average travel distance for the target installation, including door times (s)

GENERAL INTERACTION MODEL



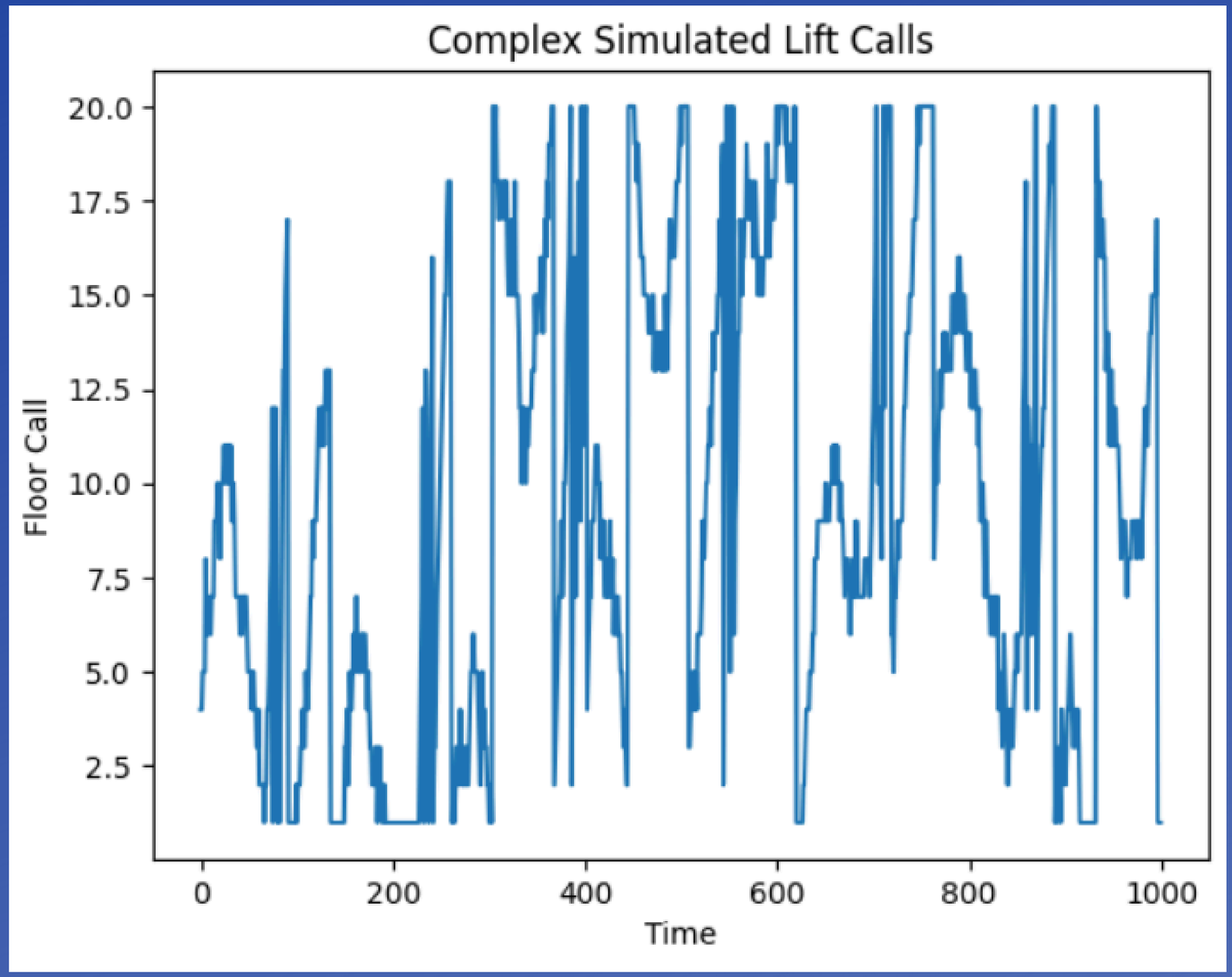
SIMULATION MODEL

- Have developed Generic and Specific lift models using python classes
- Modularization of the tasks for research speed-up
- Tested and confirmed the proper working of each module



INITIAL EXPERIMENTATION
SYNTHESIZED 1000 LIFT CALLS (SINGLE LIFT, 20 FLOORS) - USED FOR INITIAL ANALYSIS

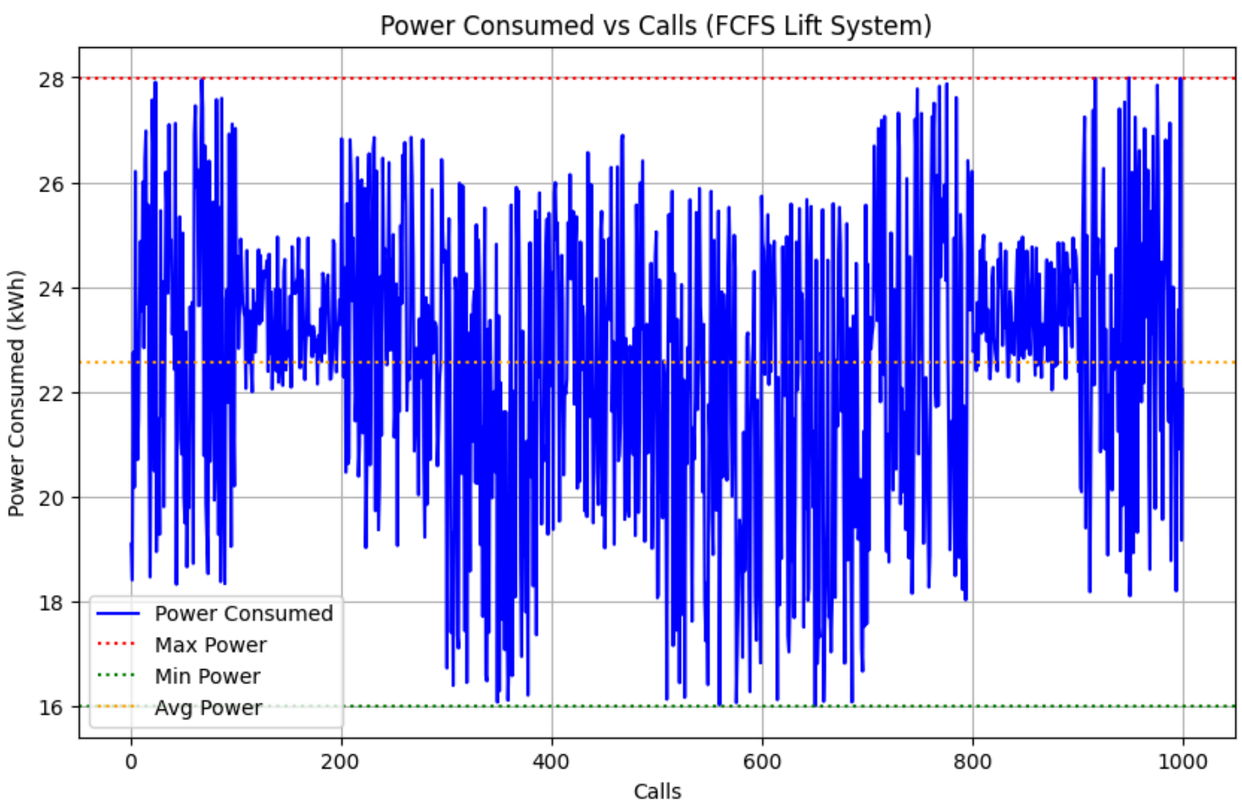
- Random floor calls (0-20) were generated over time (0-1000).
- Floor selection was randomized using uniform/normal distributions.
- Time intervals between calls were varied to simulate natural behavior.
- Periodic spikes in lift usage were added.
- Data was smoothed using moving averages to resemble real-world scenarios.

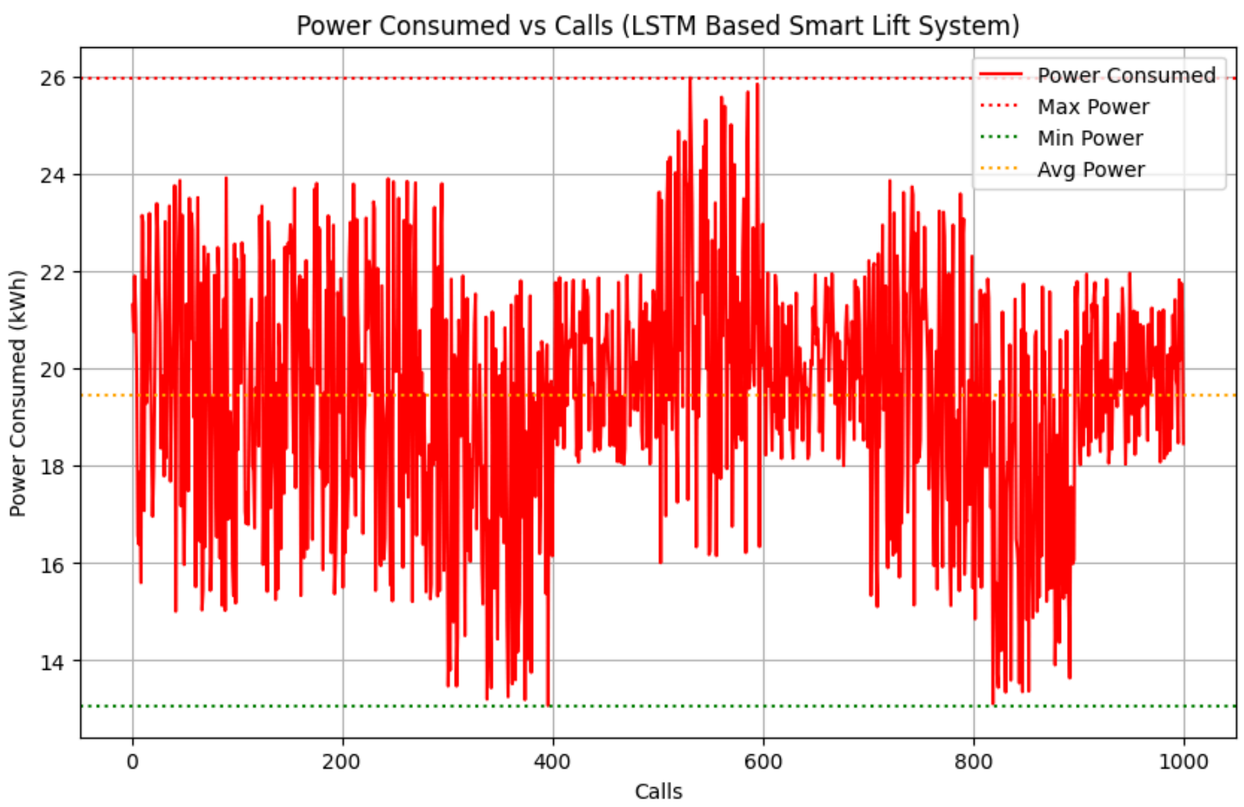


INITIAL EXPERIMENTATION (LSTM BASED SMART LIFT SYSTEM MODEL)

- The LSTM model was built with 2 hidden layers, each containing 64 units.
- The input was a sequence of previous floor calls, with a 20-step window used to predict the next call.
- The model achieved an MSE of 4.215 after training on simulated lift call data.

SIMULATION RESULTS - COMPARING POWER CONSUMED BY EACH MODEL

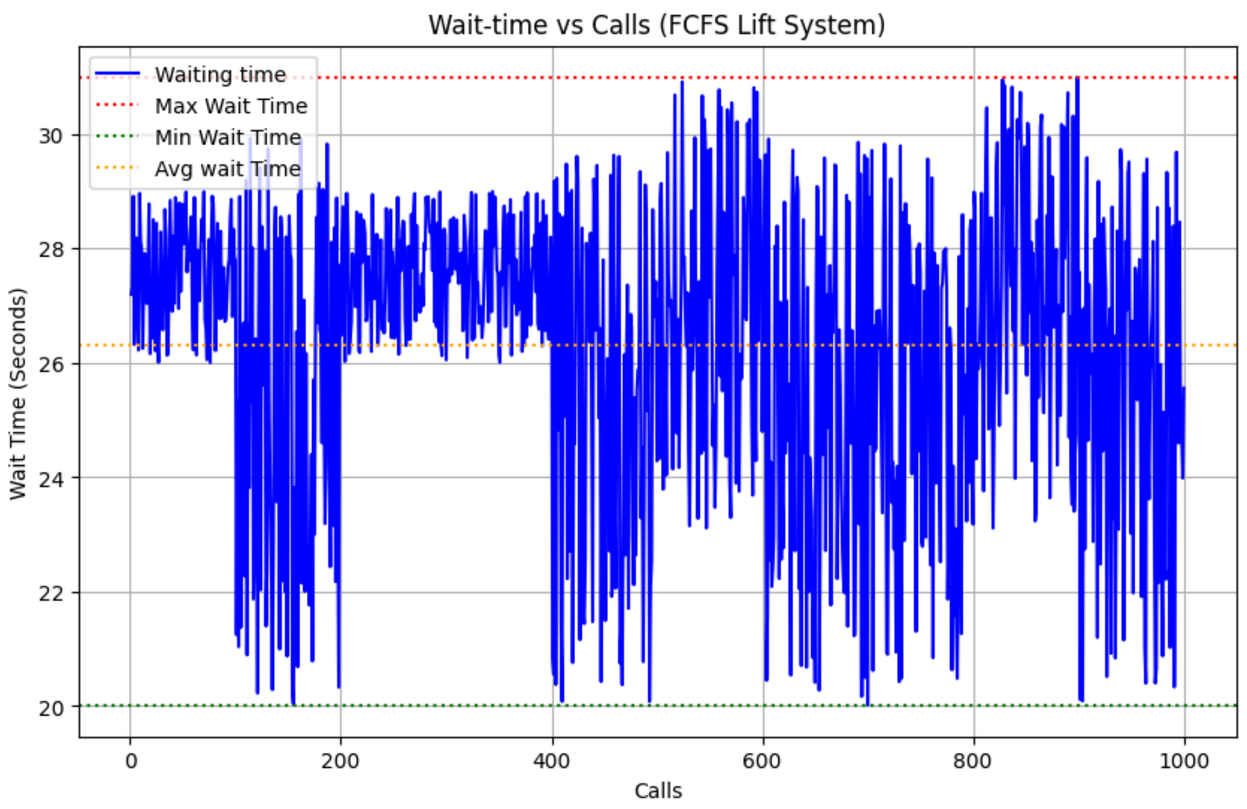


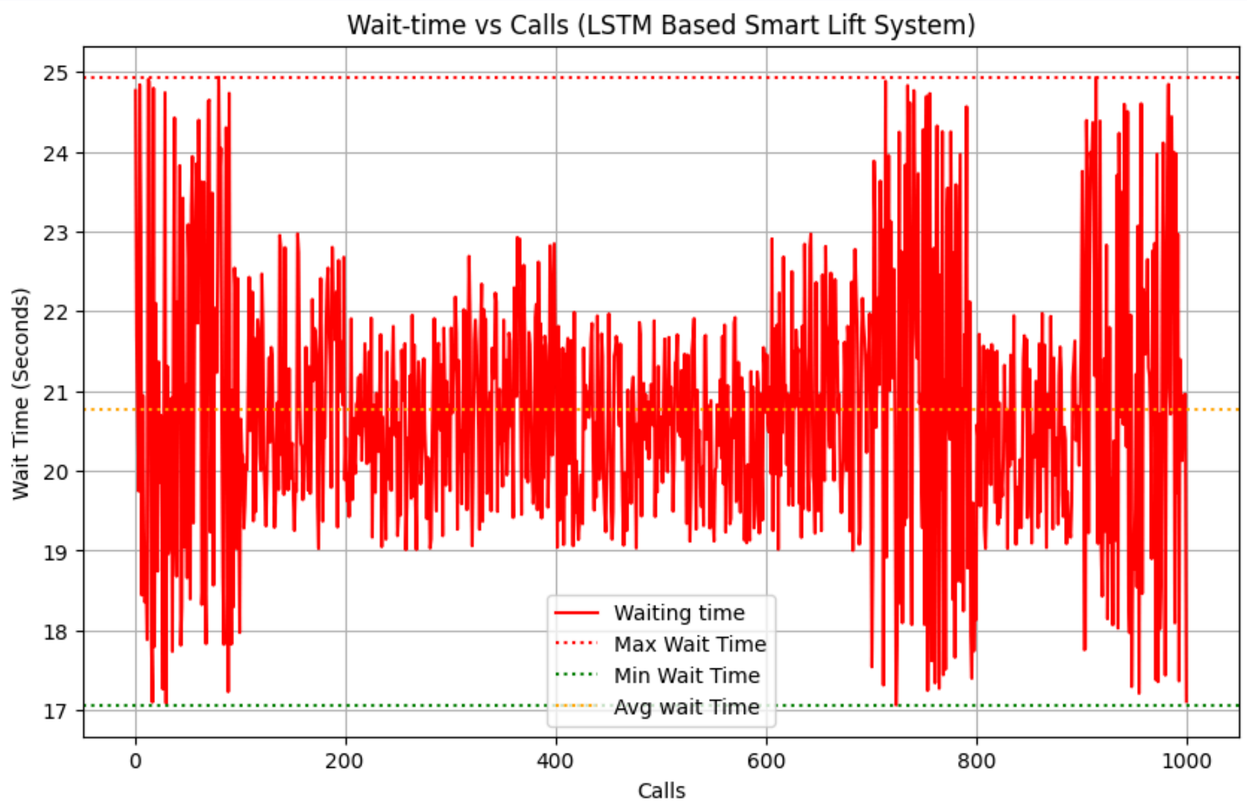


AVERAGE: 22.57 KW
MINIMUM: 16 KW
MAXIMUM: 28 KW

AVERAGE: 19.46 KW
MINIMUM: 13 KW
MAXIMUM: 26 KW

SIMULATION RESULTS - COMPARING WAIT TIME IN EACH MODEL





AVERAGE: 26 SECONDS
MINIMUM: 20 SECONDS
MAXIMUM: 31 SECONDS

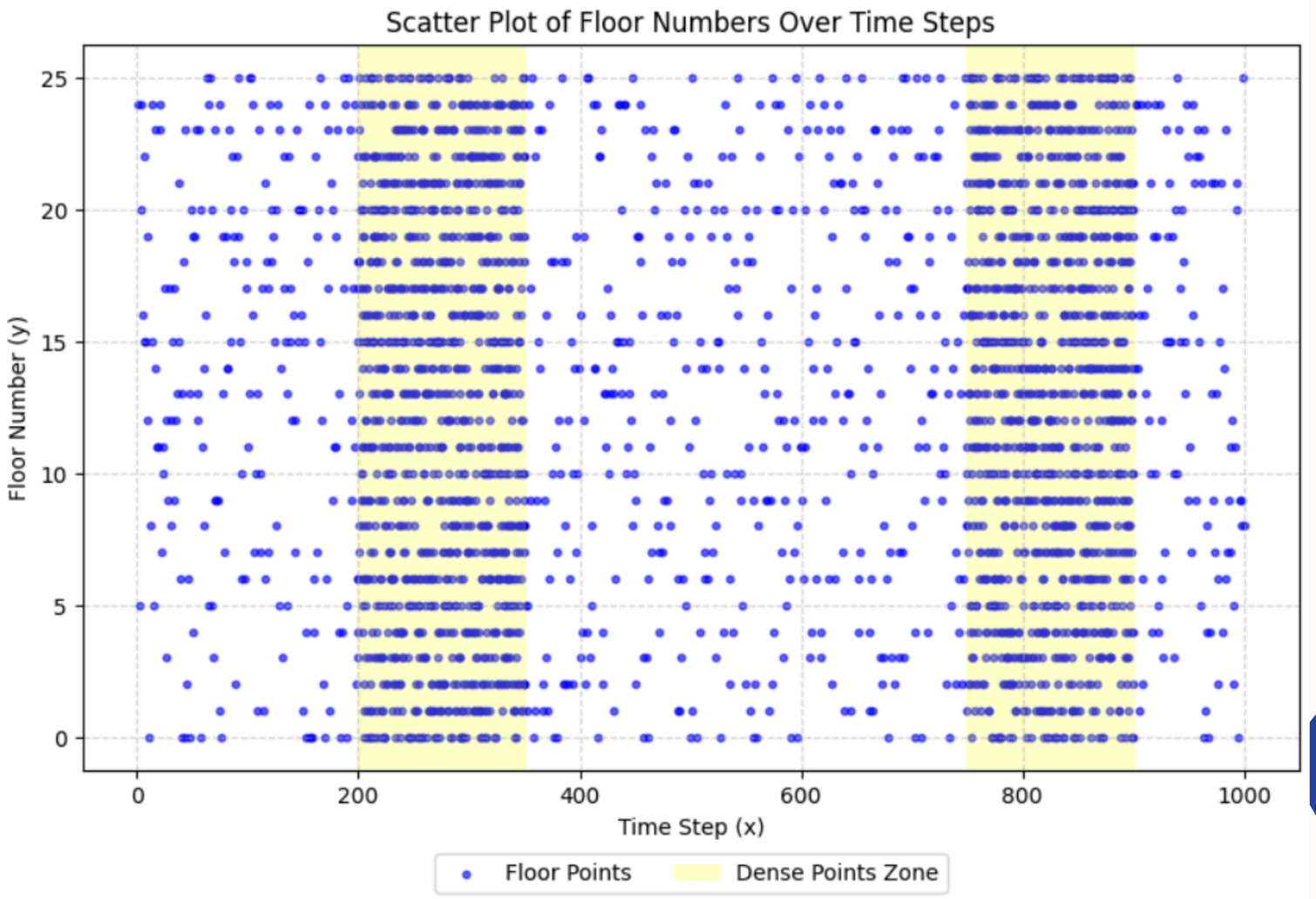
AVERAGE: 20 SECONDS
MINIMUM: 17 SECONDS
MAXIMUM: 25 SECONDS

APPROACHING MULTI-LIFT SYSTEMS

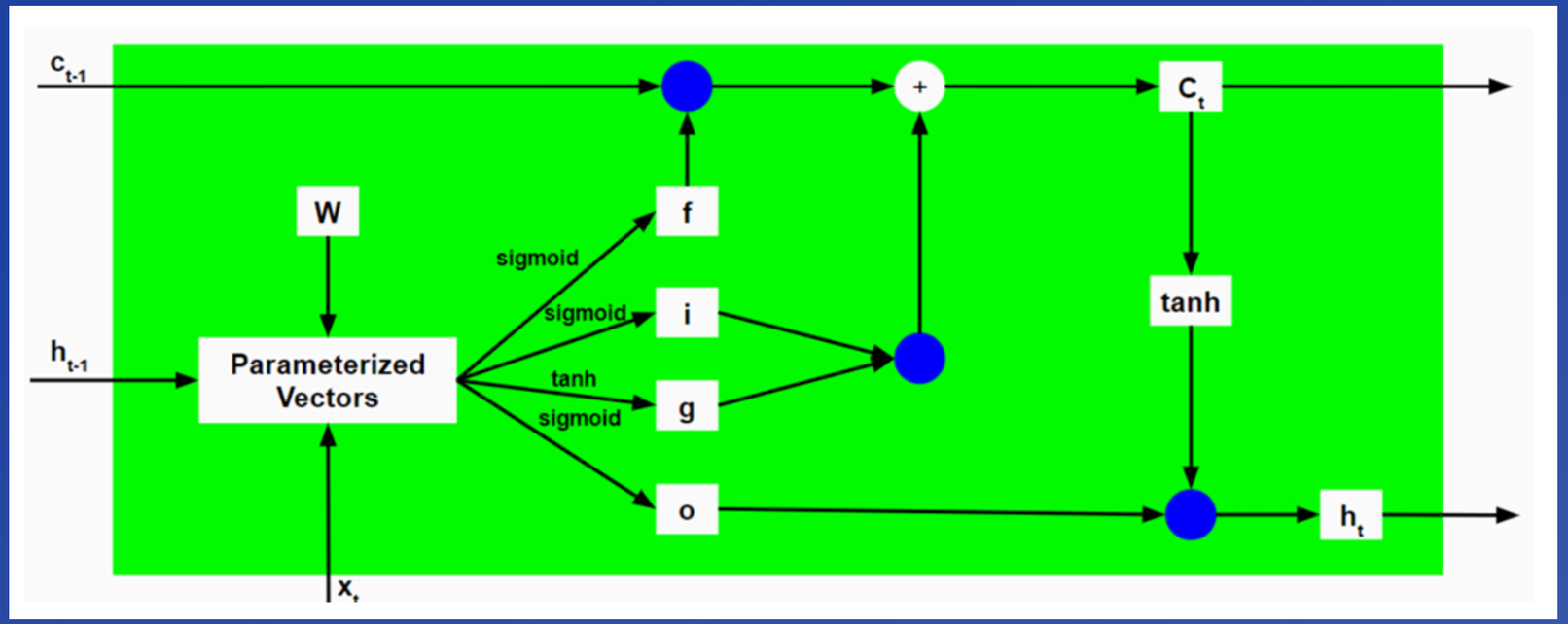
- Predict Peak Usage: LSTM models predict high-demand periods, optimizing lift availability and reducing energy waste.
- Dynamic Lift Scheduling: Predictive models ensure only necessary lifts operate, conserving energy during off-peak times.
- Optimize Load Distribution: LSTM forecasts help balance lift traffic, preventing energy waste from overloading.
- Energy-Efficient Routing: LSTM predicts the most likely next stops, minimizing unnecessary travel and saving energy.

DATA SYNTHESIS TO REFLECT PEAK HOURS

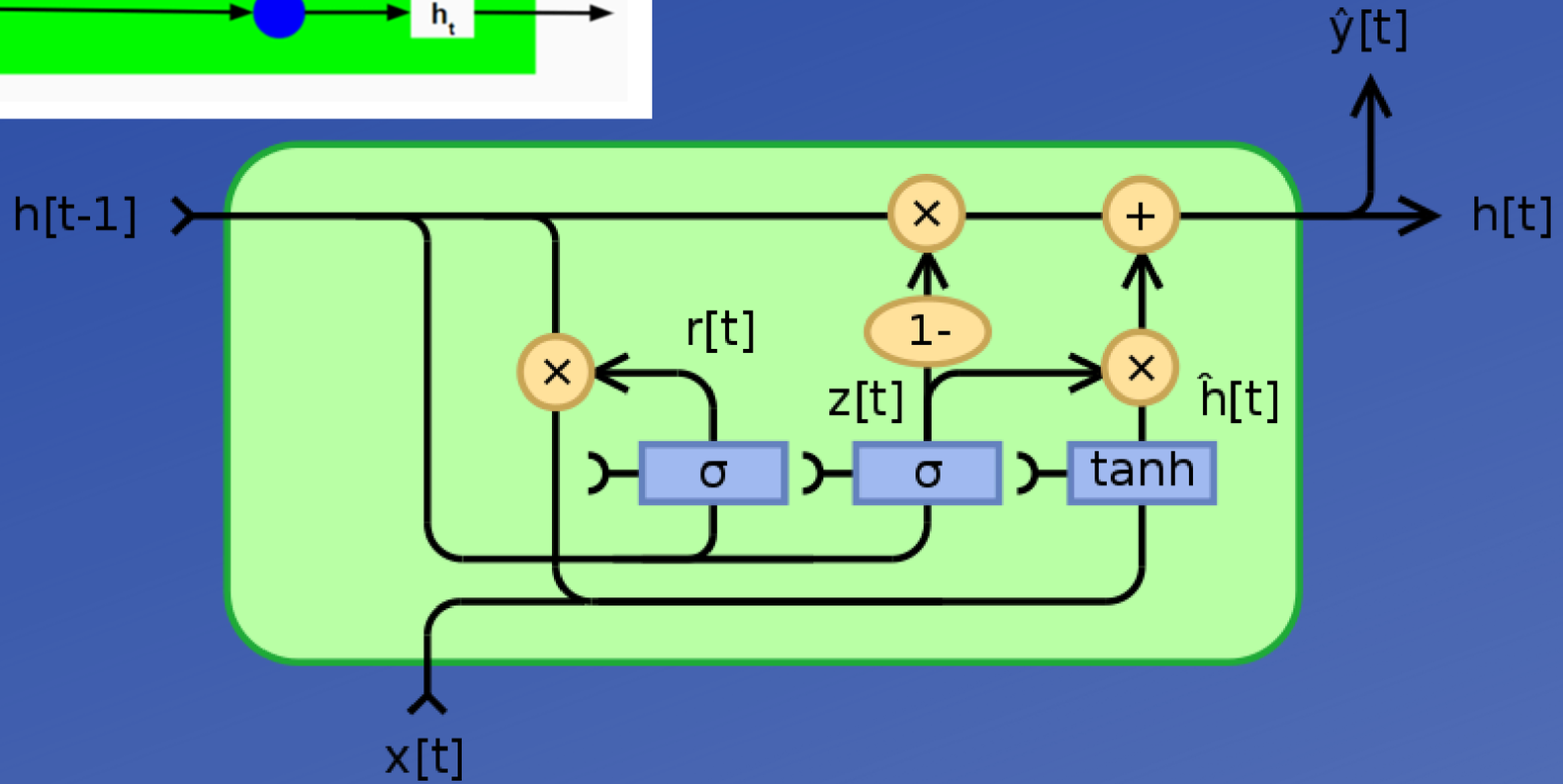
- Followed a probabilistic approach to generate data
- Peak hours decided beforehand
- Multiple random selections at peak call ranges, while single selection at non-peak ranges



PROPOSED MACHINE LEARNING MODEL ARCHITECTURES

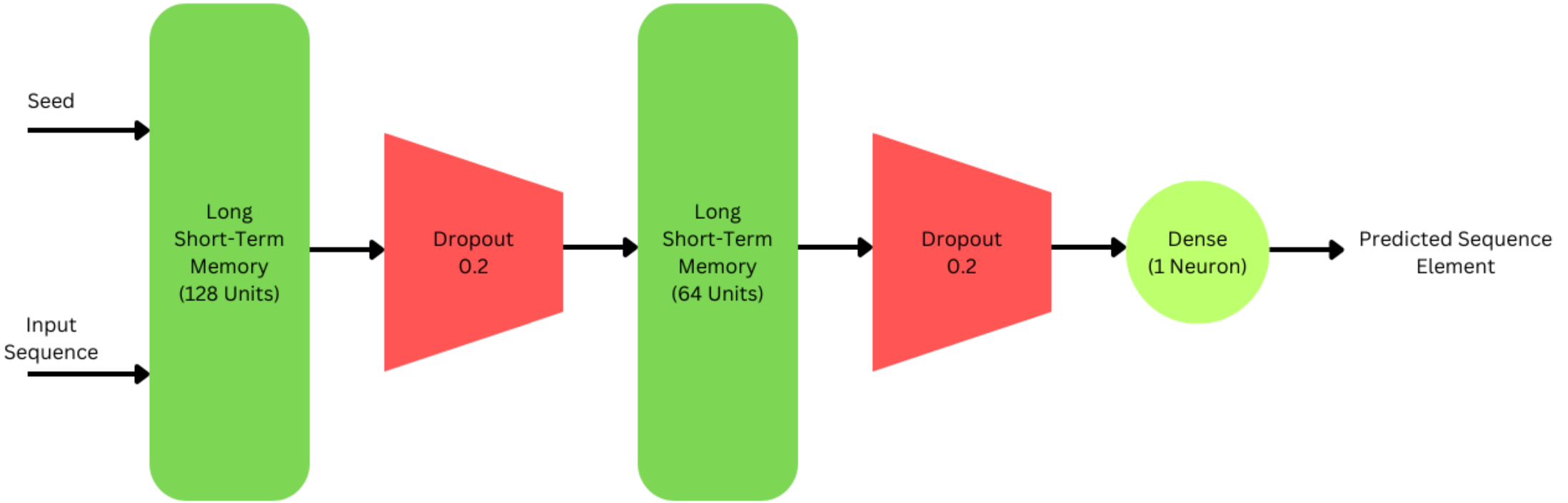


A SINGLE LSTM UNIT

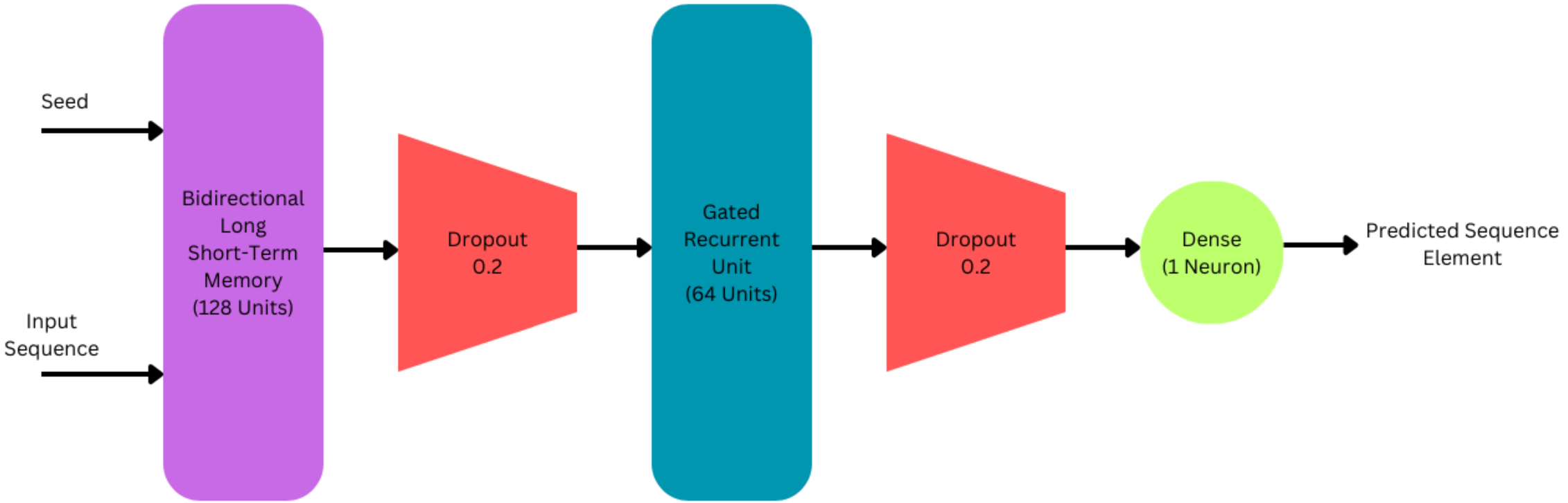


A SINGLE GRU UNIT

PROPOSED MACHINE LEARNING MODELS
1. PURE LSTM ARCHITECTURE

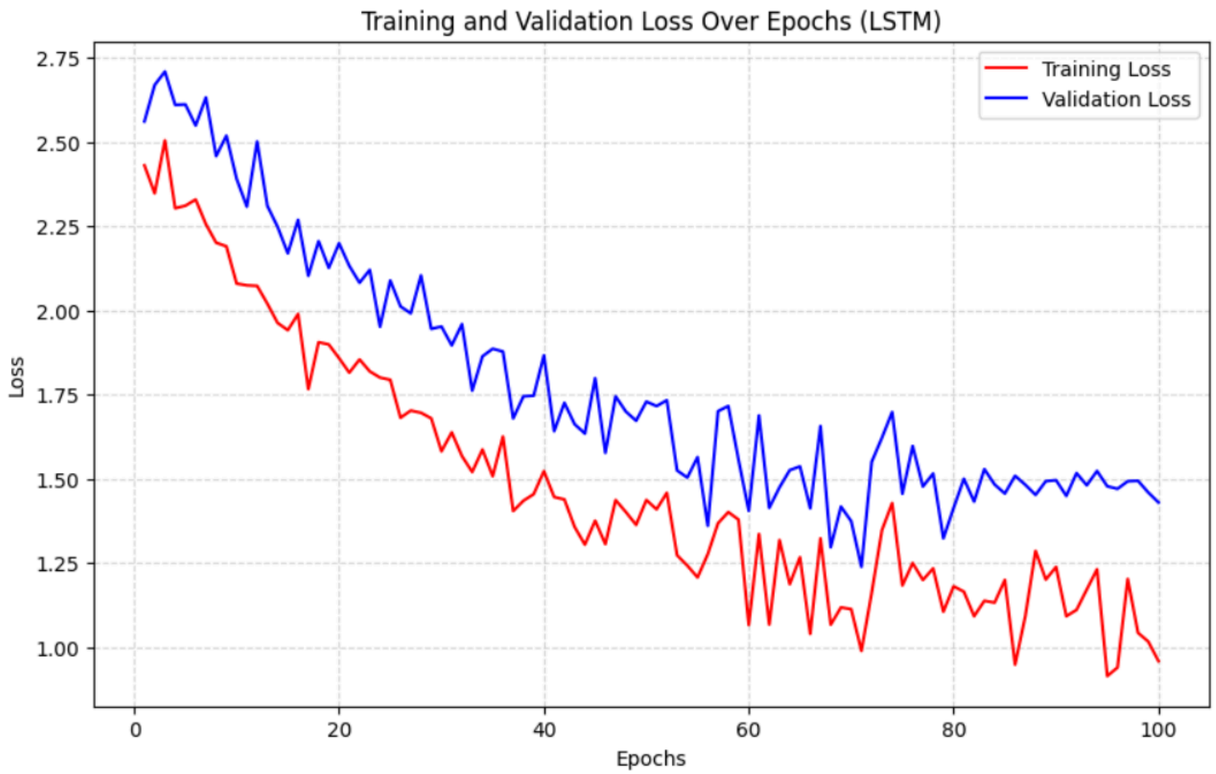


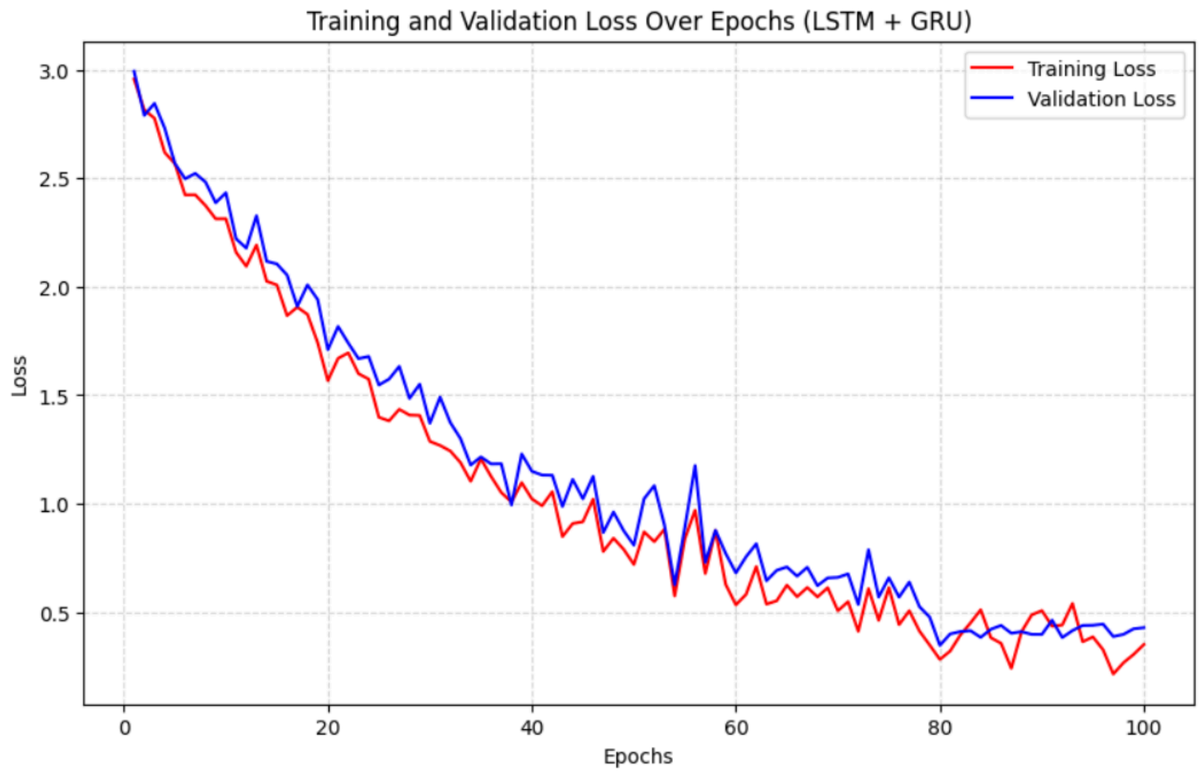
2. BIDIRECTIONAL LSTM + GRU ARCHITECTURE



RESULTS...

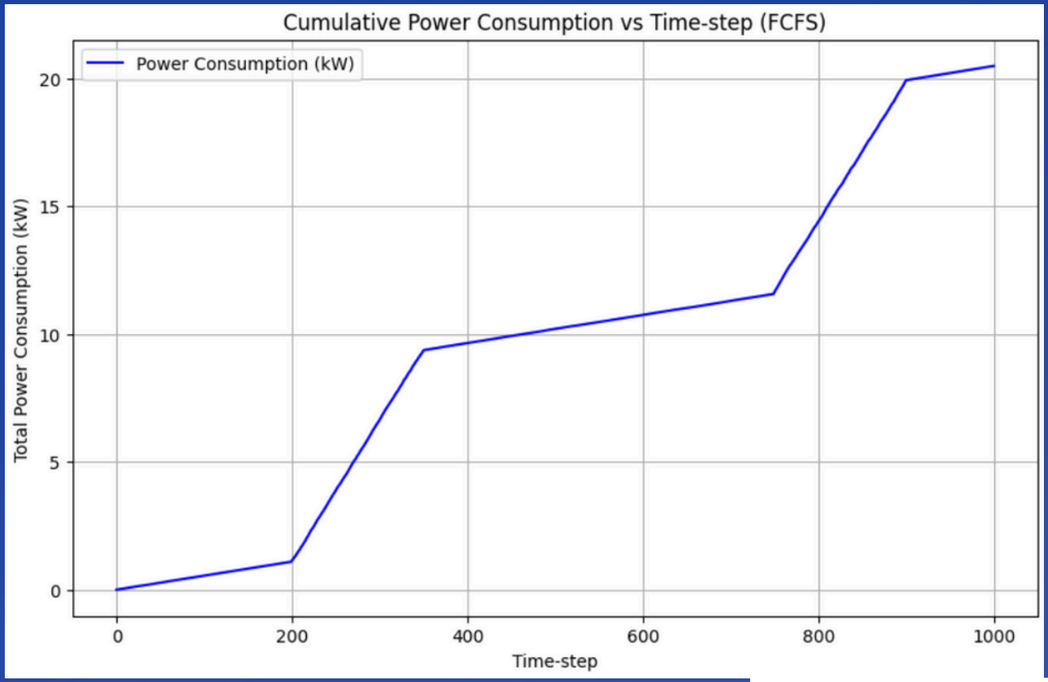
TRAINING HISTORY OF THE MODELS





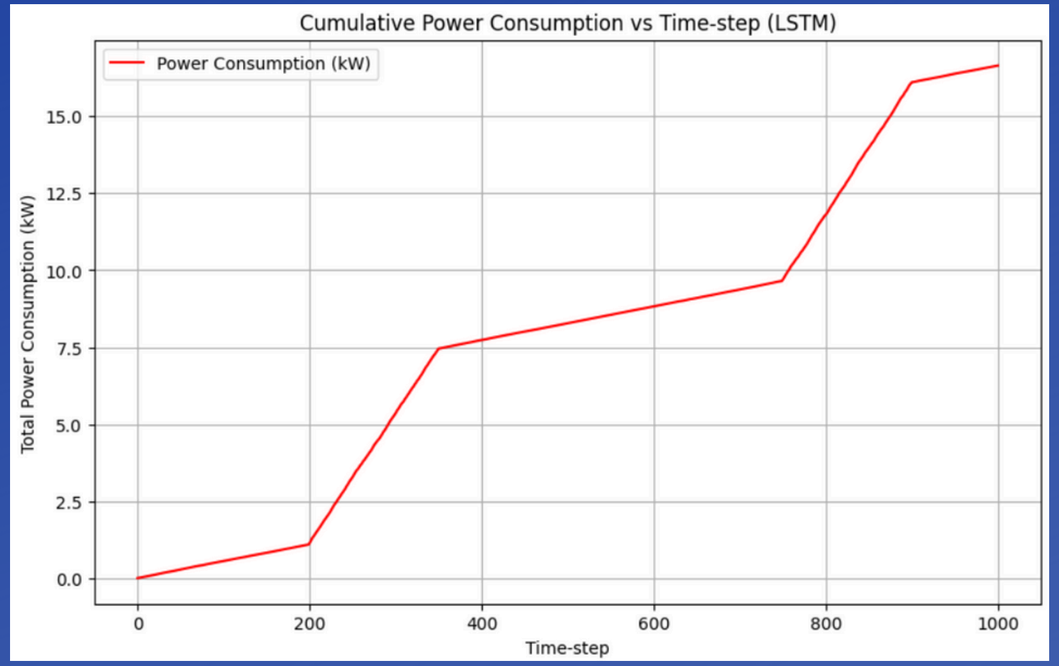
- Overall loss-
  - LSTM: Around 1.4
  - Bi-LSTM + GRU: Around 0.5

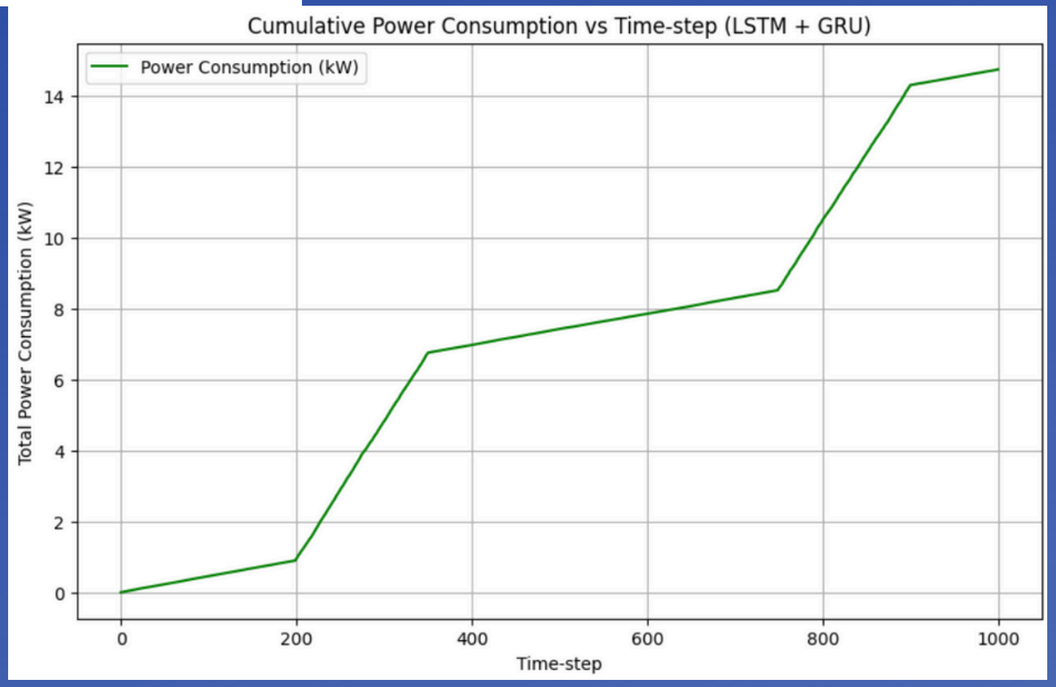
CUMULATIVE POWER CONSUMPTION EVOLUTION



FCFS
(20 KW)

LSTM +
GRU
(14 KW)





LSTM
(16 KW)

TO CONCLUDE...

- Machine Learning in multi-lift system seems to be comparatively more beneficial than in single-lift system
- Proposed architectures seem to show significant reductions in overall power consumption
- Active training might be able to overpower the mentioned pre-trained architectures

FURTHER WORK

- To implement a visual interface for the simulation software (for visual proof of advantage of ML-based lift systems)
- Further exploration of varied architectural models in search of improvements
- Testing with Real data
- Attempt at building live-training models

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